

Aryan Ashraf

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Technical Skills

Languages: TypeScript, JavaScript, C#, Python, Java, C, C++, SQL

Frameworks & Libraries: React, .NET, PyTorch, TensorFlow, YOLOv8, JavaFX, Unity

Cloud & Tools: Azure (SQL Database, DevOps), CI/CD, Docker, Git, GitHub, REST APIs, Firebase/Firestore (NoSQL)

Concepts & Practices: Unit & Integration Testing, Code Review, Data Structures & Algorithms, Object-Oriented Programming, LLM/Agent Systems, RAG

Experience

Brock Solutions

Apr. 2026 – Dec. 2026

Software Developer Intern (TypeScript · React · C# · .NET · LLM Agents)

Kitchener, ON

- Developed 10 REST API endpoints in C#/.NET with token-based authentication and account-scoped authorization, serving 10+ airline clients through a customer-facing portal on Azure SQL Database.
- Delivered a full-stack support module in React (TypeScript) and .NET with cascading, context-aware dropdowns that streamlined case intake and reduced data-entry errors.
- Built REST API integrations with ServiceNow, syncing case and support ticket data for client users.
- Authored 40 unit and integration tests across 12 endpoints covering authentication, access control, and entity validation, run in Azure DevOps CI/CD pipelines to catch regressions.
- Architected a multi-agent LLM system with retrieval-augmented generation (RAG), an orchestrator directing 5 subagents that retrieve context across 4 internal repositories to automate cross-repo analysis and planning.

University of Guelph

Apr. 2025 – Sep. 2025

ML & Assistive Devices Research Assistant (C, C#, SQL, Firestore)

Guelph, Ontario

- Debugged and extended C# logic in Unity for a VR mirror-therapy app, managing a Firestore backend that stored, synced, and retrieved patient movement and session data across devices
- Maintained Arduino (C) firmware for an arthritis rehabilitation glove, integrating sensors, real-time control logic, and breadboard wiring; validated stable operation across 20+ participants
- Analyzed 100+ temperature sensor readings for a diabetic sock project in SQL and Excel, building trend visualizations and applying Isolation Forest anomaly detection to flag abnormal patterns

Projects

University Management System | Java, JavaFX, SQL, Excel

- Designed a role-based university management system with 2 access levels (admin, user) across 5 entity types: subjects, courses, students, faculty, and events, co-designing system architecture
- Implemented authentication, scheduling, and reporting in Java and JavaFX, integrating a 10-table SQL schema and Excel import for efficient storage, updates, and usability

Real-Time ASL Detection Program | Python, PyTorch, TensorFlow, YOLOv8, Labellmg, Jupyter

- Built a real-time American Sign Language detection system in Python with PyTorch, TensorFlow, and YOLOv8, classifying 10 signs at 85%+ accuracy through a live video interface
- Annotated and managed a 1,500-image dataset with Labellmg, creating 80/20 train/test splits for model training and validation

Education

University of Guelph

Expected Apr. 2028

Bachelor of Engineering in Computer Engineering (Honours)

Guelph, Ontario

- **Relevant Coursework:** Data Structures and Algorithms (C), Digital Systems Design, Programming for Engineers (Java, C++)

Extracurricular

- **Islamic Relief at UofG:** Created marketing content and coordinated charity events to raise awareness and funds for humanitarian causes
- **Focus On Engineering Guelph:** Built and maintained a textbook rental website and helped organize engineering competitions